NarrowTech GeoSolution
narrowtechgeosolution.org

Crop Health Intelligence Platform

NarrowTech GS - Platform Guide

For Agricultural Agencies, Financial Institutions & Investors

What is NarrowTech GS?

A satellite-powered crop monitoring platform that gives farmers and agricultural agencies crop health reports from free ESA Sentinel-2 imagery - NDVI, water stress, soil moisture and in-field stress zones, using only images with 20% or less cloud over the field - delivered as PDF reports and WhatsApp messages.

Version 2.1 - October 2026

Table of Contents

1	**What is NarrowTech GS?**	Overview, mission, and value proposition
2	**The Problem We Solve**	Why traditional crop monitoring falls short
3	**How the Platform Works**	Step-by-step user journey
4	**Key Features**	Core capabilities explained simply
5	**The Science Behind It**	Satellite indices explained for non-scientists
6	**Stress Zones**	Finding the problem areas inside a field
7	**What Users See**	Platform screens and reports
8	**Notifications and Alerts**	PDF reports, WhatsApp messages, alert rules
9	**Technology Architecture**	How the system is built
10	**Data and Privacy**	How user data is handled
11	**Who is it For?**	Target users and use cases
12	**Business Model**	Pricing approach and scalability
13	**Technical Reference**	Key formulas and components
	Version History	What changed in versions 2.0 and 2.1

1. What is NarrowTech GS?

NarrowTech GS (GeoSolution) is a web-based crop health monitoring platform developed by NarrowTech GeoSolution. It uses satellite imagery from the European Space Agency's Sentinel-2 mission to give farmers, agricultural officers, and development agencies an accurate, timely picture of crop health - without expensive field visits or equipment.

In simple terms: Think of it as a check-up for your farm - done by satellite, delivered to your phone as a PDF report and WhatsApp message. It can also point you to the exact spots in the field that need attention.

Free	10 m	14 days	Any device
Satellite data	Native resolution	Observation window	Browser access

Mission

To democratize precision agriculture by making satellite-based crop monitoring accessible to smallholder farmers and agricultural institutions across Africa and beyond - at zero cost for the satellite data itself.

Core Value Proposition

- **No equipment needed:** Entirely satellite-based - no sensors, drones, or hardware required.
- **Real satellite data only:** Every value comes from an actual Sentinel-2 image. Nothing is simulated or estimated; when no clear image exists, the platform says so.
- **Timely alerts:** WhatsApp and PDF updates when crop health deteriorates.
- **Any field, anywhere:** Users draw, upload, or type their field boundaries.
- **Easy to use:** Designed for non-technical users - no GIS knowledge needed.
- **Actionable insights:** Clear recommendations like 'Schedule irrigation within 3-5 days' and GPS pins to the worst-affected spots - not just numbers.

2. The Problem We Solve

Crop failure and water stress are among the leading causes of food insecurity and agricultural losses in sub-Saharan Africa. Most farmers and agricultural agencies lack timely, accurate information about what is happening in their fields.

Traditional scouting is slow	Walking fields to assess crop health takes days - by which time drought or disease may have caused irreversible damage.
Lab tests are expensive	Soil and crop analysis requires expensive equipment and weeks of waiting.
Satellite data is too complex	Platforms like Google Earth Engine require GIS expertise - not accessible to farmers or field officers.
No early warning system	Most farmers only discover problems when visible with the naked eye - often too late to intervene.
Weather forecasts are insufficient	Knowing it will rain does not tell you if your crops are already stressed right now.
Problems are patchy	Stress rarely affects a whole field evenly; without a map, farmers do not know where to look first.

NarrowTech GS addresses these with satellite analysis in 14-day windows, plain-language recommendations, stress-zone pins that lead you to the problem spots, and updates delivered by WhatsApp and PDF.

3. How the Platform Works

NarrowTech GS guides the user through a simple 5-step process from sign-in to receiving crop health updates.

1	**Sign In** The first account created on the platform becomes the administrator. Public sign-up is then closed: the administrator creates accounts for farmers and staff, who sign in with email and password.
2	**Add Your Field** Define the field boundary by (a) drawing it on an interactive map, (b) uploading a GeoJSON file or a zipped Shapefile, or (c) typing coordinates. Choose the region (all 16 Ghana regions), crop type and analysis date range.
3	**Run Satellite Analysis** The platform measures how cloudy your field is on every Sentinel-2 Level-2A image in the date range, split into 14-day windows. In each window it uses the clearest image with 20% or less cloud over the field, masks the remaining cloud, shadow and water pixels, and computes NDVI, MSI and soil moisture. The analysis runs in the background and the screen updates when it finishes.
4	**View Results** The dashboard shows NDVI (crop vigour), MSI (water stress) and a Health Score (0-100). Time-series charts show the trend, colour maps show the field, and stress zones mark the worst spots with GPS pins.
5	**Receive Updates** Download a PDF crop health report or send a WhatsApp summary for any field. An administrator can run a bi-weekly check that refreshes every field due for an update.

How images are chosen - the 20% cloud rule

- **Only Sentinel-2 Level-2A images** (atmospherically corrected surface reflectance) are used.
- **Cloud is measured over your field**, not over the whole 100 x 100 km satellite tile. Cloud, thin cirrus, cloud shadow and missing pixels all count as cloud.
- **An image is used only if 20% or less of the field is cloud.** The limit is fixed for every analysis.
- **One image per 14-day window:** the clearest qualifying image taken inside the window.
- **If a window has none,** the nearest qualifying image taken up to 7 days before or after the window is used, provided no other window already uses it. Otherwise the window is left empty.
- **Within a used image,** the remaining cloud, shadow and water pixels are removed, so every average, map and stress zone is built only from clear pixels inside the field.

Satellite data only. A window with no qualifying image is shown as a gap - no value is invented. If the satellite service cannot be reached, or no image with 20% or less field cloud exists in the whole date range, the analysis stops with a clear message and the field keeps its previous results.

4. Key Features

Satellite Analysis	NDVI and MSI computed from Sentinel-2 Level-2A via the Copernicus Data Space Sentinel Hub APIs.
Soil Moisture (OPTRAM)	Relative soil wetness index (0 = dry, 1 = wet) estimated per pixel with the Optical Trapezoid Model.
Stress Zones	Finds the worst-performing patches inside the field and returns up to 5 GPS pins with walking directions in Google Maps.
Dashboard	Central view of all fields with health status, NDVI, MSI, health score and status badges.
Time Series Charts	NDVI, MSI and soil moisture over time, with the image date, 14-day window and field cloud % for every point. Export as CSV or PNG.
Spatial Maps	Colour-coded NDVI and MSI maps for each image used in the time series. Cloud and shadow shown grey, water blue. Download as PNG.
Alerts System	Three severity levels: Info, Warning, Critical. Alerts can be acknowledged once addressed.
PDF Reports	A4 reports with KPI summary, NDVI/MSI gauges, trend chart, active alerts, observation log with field cloud %, and methodology.
WhatsApp Messages	Plain-text crop health summaries via the Meta WhatsApp Cloud API. Works on any WhatsApp account.
Multi-field Management	Unlimited fields. Full time series and alert log per field.
Cloud Filtering	Only images with 20% or less cloud (including shadow) over the field itself. Nearest qualifying image within 7 days fills an empty window.
Water Pixel Removal	Water pixels (scene class 6) are masked; a date on which half or more of the field is water is flagged and left out of stress analysis.

5. The Science Behind It (Simply Explained)

NarrowTech GS uses three satellite measurements to assess crop health. All are calculated from the way plants and soil reflect light - which satellites can detect even when it is invisible to the human eye. NDVI and MSI drive the Health Score; soil moisture adds supporting detail. All values are averaged over the clear pixels inside the field only.

NDVI - Normalized Difference Vegetation Index

In simple terms: NDVI measures how green and dense the vegetation is. High NDVI = healthy, actively growing plants. Low NDVI = bare soil, dry crops, or sick plants.

NDVI Range	Meaning	Action
0.6 - 1.0	Excellent - healthy, dense crops	Maintain current practices
0.4 - 0.6	Good - crops growing well	Continue routine monitoring
0.25 - 0.4	Moderate - some stress	Investigate cause
0.1 - 0.25	Poor - significant stress	Apply fertiliser, check for pests
Below 0.1	Critical - bare soil or crop failure	Emergency field inspection

MSI - Moisture Stress Index

In simple terms: MSI measures water stress. HIGHER MSI = BAD (plant running out of water). Very low MSI may indicate open water (flood or pond).

MSI Range	Meaning	Action
Below 0.1	Open water (excluded from analysis)	Water - ignored
0.1 - 0.4	High moisture - very well watered	No action needed
0.4 - 0.6	Adequate moisture	Continue monitoring
0.6 - 1.0	Mild water stress	Consider irrigation planning
1.0 - 1.2	Moderate drought stress	Schedule irrigation within 3-5 days
Above 1.2	Severe drought - URGENT	Irrigate immediately

Soil Moisture - OPTRAM Index

In simple terms: The Optical Trapezoid Model (OPTRAM) compares how much shortwave-infrared light the ground reflects with how green it is, and places every pixel between the driest and wettest conditions seen in the field. The result is a wetness index from 0 (dry) to 1 (wet).

Because the dry and wet limits are calibrated from the images in each analysis, the index is **relative**: it shows whether the field is wetter or drier than at other dates in the same analysis, not an absolute water content. It is shown on the time-series chart and in the CSV export; it does not affect the Health Score or alerts. Dates without enough valid pixels show no soil-moisture value.

Health Score (0–100)

A composite score combining NDVI (70% weight) and MSI (30% weight), calculated from the most recent valid observation.

Health Score	Status	What happens
70 - 100	Healthy	Green badge
45 - 69	Monitor	Yellow badge
Below 45	Alert	Red badge; field status set to Alert

6. Stress Zones - Finding the Problem Areas

A field average can hide a problem: one dry corner or a pest patch can be missed when the rest of the field is healthy. Stress Zones point the farmer to the exact spots that need attention.

In simple terms: The platform compares every part of the field with the field's own average, groups the worst neighbouring spots together, and drops a labelled pin (Zone A, B, C…) on each group. Tapping a pin opens walking directions in Google Maps.

How it works

- The user picks an image date and an index (NDVI or MSI) on the Spatial Maps page. Only the images used in the time series (20% or less field cloud) can be chosen.
- The platform requests a 40 x 40 grid of index values over the field from exactly that image.
- Grid cells outside the drawn field boundary, and cells covered by cloud, cloud shadow or water, are removed.
- Each remaining cell is compared with the field average. Cells that are clearly worse (more than 0.12 below average on a 0-1 scale; above average for MSI) are treated as stressed.
- Touching stressed cells are grouped into zones (DBSCAN clustering). Each zone's pin sits at its severity-weighted centre.
- Zones are ranked by severity - Critical (more than 0.25 worse than average) or Warning - and up to 5 are shown.

If the field looks uniform, the platform reports that there are no distinct stress zones for that date rather than forcing pins onto healthy ground.

7. What Users See

Platform URL: **https://app.narrowtechgeosolution.org** - accessible from any browser on desktop or mobile. No app installation required.

Sign In

Secure sign-in screen with email and password. The first account becomes the administrator; the administrator creates further accounts.

Dashboard

Overview of all fields as cards. Each card shows field name, region, crop type, NDVI, MSI, health score and status badge, with buttons to re-analyze, view the time series, download the PDF report, or send a WhatsApp update. A banner warns if the satellite service cannot be reached.

Field Analysis

Form to add a field: draw the boundary on the map, upload a GeoJSON or zipped Shapefile, or type coordinates. KML and GeoPackage files should first be exported to GeoJSON (the page explains how in QGIS, ArcGIS and Google Earth). Set region, crop type and date range. The 20% cloud rule is applied automatically.

Time Series

Line chart of NDVI, MSI and soil moisture over time, one point per 14-day window. The observation table lists, for each window, the image date used, the field cloud %, and whether it is the window's own image or the nearest one borrowed from up to 7 days away. Windows with no qualifying image are marked 'No image 20% or less cloud'. Export as CSV or PNG.

Spatial Maps

Colour-coded NDVI (red to green) and MSI (blue = moist, red = stressed) maps for every image used in the time series, labelled with its field cloud %. Cloud and cloud shadow show grey, water blue. Stress-zone pins with Google Maps directions. Download maps as PNG.

Alerts

Log of all alerts across all fields, colour-coded by severity (Info / Warning / Critical). Acknowledge alerts once addressed.

8. Notifications and Alerts

NarrowTech GS delivers crop health updates in two formats: a downloadable PDF report and a WhatsApp message.

PDF Report Contents

- Field name, region, crop type, and analysis period
- Health Score, NDVI and MSI with status labels and colour gauges
- Actionable recommendation in plain English (e.g. 'Schedule irrigation within 3-5 days')
- Time-series chart of NDVI and MSI
- Active alerts
- Table of the last 12 valid satellite observations with dates and data sources
- Satellite data methodology and source notes
- No sensitive information (no API keys, user emails, or passwords)

WhatsApp Message

A plain-text message sent through the Meta WhatsApp Cloud API. It includes the results, a recommendation and up to three active alerts. Sample:

```
NarrowTech GS - Crop Health Update
Field: Tono Irrigation Site
Region: Upper East
Crop: Rice

Satellite Results (Sentinel-2)
- NDVI: 0.826  ->  Excellent
- MSI:  0.193  ->  High moisture
- Health Score: 89/100

Recommendation
Crop is healthy. Maintain current practices.

Period: 2026-03-01 -> 2026-06-14
Analyzed: 14/06/2026
-- NarrowTech GeoSolution
narrowtechgeosolution.org
```

Delivery today

- **On demand:** any signed-in user can download a field's PDF report or send its WhatsApp update from the dashboard.
- **Bi-weekly check:** an administrator can refresh every field that has not been updated in the last 14 days with one button. Fields with no usable satellite image are listed and keep their previous results.
- **Pilot setup:** WhatsApp messages currently go to one configured phone number. Per-farmer phone numbers and fully automatic 14-day scheduling are planned.

Alert Rules

Condition	Severity	Message
NDVI below 0.2	Critical	Possible crop failure or bare soil
NDVI 0.2 - 0.35	Warning	Low vegetation density - check for nutrient or pest stress
MSI above 1.2	Critical	Severe water stress - immediate irrigation recommended
MSI 1.0 - 1.2	Warning	Moderate water stress - consider supplemental irrigation
NDVI above 0.7 and MSI below 0.6	Info	Excellent crop health

9. Technology Architecture

NarrowTech GS is a modern web application with a clear separation between the user interface (frontend), the data-processing engine (backend), and external data sources.

Layer	Technology	Role
Frontend (UI)	React + TypeScript + Tailwind CSS	User interface - runs in any browser, mobile-friendly
Maps	Leaflet.js	Interactive map for drawing and viewing field boundaries
File import	shpjs	Reads zipped Shapefiles in the browser
Charts	Recharts	Time-series NDVI / MSI / soil-moisture charts
Backend (API)	Node.js + Express.js	Processes requests, calls satellite APIs, runs analyses as background jobs, generates reports
Database	SQLite (sql.js) + Drizzle ORM	Stores fields, time series, alerts, settings and users; saved to a dedicated storage disk
Satellite Data	Copernicus Data Space Sentinel Hub (Statistics + Process APIs)	Sentinel-2 Level-2A statistics, raster data and map images
Authentication	bcrypt + session cookies	Secure sign-in with hashed passwords, 7-day sessions
PDF Reports	PDFKit (Node.js)	Generates A4 crop health reports
WhatsApp	Meta WhatsApp Cloud API	Crop health messages to WhatsApp
Hosting	Render.com + custom domain	HTTPS at app.narrowtechgeosolution.org

10. Data and Privacy

NarrowTech GS is designed with data minimization and user privacy in mind.

What is stored	Field boundaries (GPS), crop type, region, analysis results and time series (NDVI, MSI, soil moisture, field cloud %, health score), alert history, user name, email, role, and hashed password.
What is NOT stored	Raw satellite imagery, financial information, personal ID beyond name/email, or any sensitive government data.
Satellite data	All imagery from ESA Copernicus - publicly available free of charge. No proprietary data used.
Password security	All passwords hashed with bcrypt (10 rounds). Plain-text passwords never stored.
Access control	Public sign-up is closed; only the administrator can create accounts.
PDF reports	Reports contain no API credentials, server paths, or user emails - only field results.
Credentials	Satellite and WhatsApp keys are kept in server environment settings, never in reports or messages.

11. Who is NarrowTech GS For?

User Type	How They Use It	Key Benefit
Smallholder Farmers	Receive WhatsApp updates and walk to stress-zone pins	Early warning of drought or crop failure - time to act
Agricultural Extension Officers	Monitor multiple farmer fields from one dashboard	Remote oversight of farming districts with fewer field visits
Irrigation Schemes	Track water stress and soil moisture across irrigated plots	Irrigation scheduling based on actual crop data
NGOs & Development Agencies	Assess project impact across farmer beneficiaries	Evidence-based reporting on agricultural interventions
Financial Institutions	Verify crop health for loan or insurance assessments	Objective satellite-based crop status without physical inspection
Government Ministries	Early warning for food security at district level	Scalable monitoring of many farms at low cost
Research Institutions	Analyse crop health trends over time and geography	Longitudinal data with a standardised satellite methodology

12. Business Model

NarrowTech GS is positioned as a Software-as-a-Service (SaaS) platform. The satellite data (Copernicus Sentinel-2) is free, keeping costs low and enabling affordable pricing for African markets.

Model	Description	Target
Freemium	Free tier: 1 field, basic dashboard. Paid: unlimited fields, WhatsApp, PDF, maps, stress zones.	Individual farmers, cooperatives
Institutional License	Annual subscription for NGOs/government to monitor hundreds of fields.	Government bodies, development agencies
API Access	Enterprise clients access the analysis engine via API for system integration.	AgriTech companies, banks, insurers
Grant Funding	Platform deployed under donor-funded projects as a monitoring/verification tool.	World Bank, USAID, AfDB
Pay-per-Report	One-off PDF or WhatsApp reports for specific fields.	Loan officers, insurance assessors

Competitive advantage: Copernicus Sentinel-2 data is free (EU-funded). This gives NarrowTech GS a structural cost advantage over platforms using commercial satellites. Main cost drivers are hosting and WhatsApp messaging - both low at scale.

13. Technical Reference

Key technical details for developers and technical reviewers. Source code: github.com/Ramson94/Narrotech--gs (server/routes.ts holds the analysis engine).

NDVI	(B08 - B04) / (B08 + B04). B08 = near-infrared, B04 = red. Range -1 to 1.
MSI	B11 / B8A. B11 = SWIR (1610 nm), B8A = narrow NIR (865 nm). Higher = more stress.
OPTRAM soil moisture	STR = (1 - B11)^2 / (2 x B11). Per-pixel NDVI and STR (about 20 m) pooled across all dates; NDVI split into 35 bins; dry and wet edges fitted by linear regression through the 2nd and 98th STR percentiles of each bin. W = (STR - STR_dry) / (STR_wet - STR_dry), kept for 0 to 1.05 and averaged per date. Pixels with B11 below 0.05 or STR above 5 are discarded; the model is not used if the fitted edges fail physical checks.
Field cloud %	Statistics API (daily intervals) over the field polygon at about 10 m (32-256 px per side). Field cloud % = share of field pixels in SCL class 3 (cloud shadow), 8, 9 (cloud), 10 (cirrus), 0/1 (no data/defective) or outside the swath. No tile-level cloud filter is used.
Image selection	Images with field cloud of 20% or less qualify. Each 14-day window takes its lowest-cloud qualifying image (ties: closest to the window's middle). Empty windows then take the nearest unused qualifying image up to 7 days outside the window (ties: lower cloud). Others are stored as gaps. Code: server/imagery.ts.
Pixel averaging	NDVI and MSI are averaged only over clear land pixels (not cloud, shadow, cirrus, no-data or water, SCL 6) inside the polygon: mean(value x clear) / mean(clear). A date with 50% or more water is flagged as water.
No-data handling	Windows without a usable image are stored as excluded gaps with no values. If the satellite service is unreachable or no window has data, the analysis fails with a clear message. No simulated or estimated values are produced.
Health Score	NDVI score x 0.7 + MSI score x 0.3. NDVI score = (NDVI - 0.05) / 0.9 x 100, clipped to 0-100. MSI score: below 0.4 = 95, 0.4-0.6 = 80, 0.6-1.0 = 60, 1.0-1.2 = 35, above 1.2 = 10, missing = 50. Bands: 70+ Healthy, 45-69 Monitor, below 45 Alert.
Spatial maps	Only dates that are qualifying images in the field's time series. Process API renders a 256 x 256 PNG of exactly that day's image (~500 m buffer). Cloud, cirrus, shadow and no-data pixels grey; water blue.
Stress zones	Same qualifying-date rule. 40 x 40 grid of NDVI or MSI values (8-bit) inside the field polygon, cloud/shadow/water cells removed; cells more than 0.12 worse than the field mean clustered with DBSCAN (8-connected neighbours); severity-weighted pin centres; Critical above 0.25; up to 5 zones (maximum 10).
Background analysis	POST /api/fields/:id/analyze returns a job ID immediately; the app polls /api/analyze-jobs/:jobId so long analyses do not hit hosting timeouts.
Bi-weekly check	POST /api/notify/check (administrator only) re-analyzes every field whose last update is 14 or more days old, using the past 14 days of imagery (plus the 7-day fallback), and adds the result to the field's existing time series.
Authentication	bcrypt (10 rounds) password hashing; in-memory session store; 7-day secure session cookie. Users sign in again after a server restart.
Database tables	fields, alerts, app_settings, users. SQLite via sql.js (WebAssembly) and Drizzle ORM, written to /data/data.db.
Environment settings	COPERNICUS_USER_ID, COPERNICUS_ACCOUNT_ID, META_WA_TOKEN, META_WA_PHONE_ID, WHATSAPP_NOTIFY_TO, SESSION_SECRET, PORT.

Version History

Version 2.1 (October 2026)

Area	Change
Cloud rule	Only Sentinel-2 L2A images with 20% or less cloud over the field itself are used. Cloud is no longer judged from the whole satellite tile, and the automatic retry at 50% was removed. The limit is fixed at 20%.
Cloud shadows	Counted as cloud for the 20% rule and masked everywhere, including maps and stress zones.
Empty windows	Use the nearest qualifying image up to 7 days outside the window; otherwise a gap.
Maps & stress zones	Built only from the exact images used in the time series.
EVI removed	EVI was removed from analysis, alerts, maps, charts, reports and messages; NDVI covers vegetation vigour.
Field averages	Calculated over the drawn field shape only, not its surrounding rectangle.
Time series table	Shows the real image date, its 14-day window, and the cloud % over the field.
Bi-weekly check	Adds new images to a field's history instead of replacing it.

Version 2.0 (October 2026)

Area	Change
Soil moisture	OPTRAM relative wetness index documented.
Stress Zones	In-field problem spots with GPS pins and Google Maps walking directions documented.
Satellite data only	The simulated-data fallback was removed. Gaps are shown as gaps; failures are reported clearly.
Field upload	GeoJSON and zipped Shapefiles. KML and GeoPackage must first be exported to GeoJSON.
Accounts	Public sign-up closed after the first (administrator) account.
Health Score	Alert threshold aligned at 45 across the platform.
Notifications	Updates are sent on demand or through the administrator's bi-weekly check, to one pilot WhatsApp number.